

Response to Reviewers' Comments

TNET-2025-01079 entitled "Handling Qubit Allocation and Topology Design at Scale for Distributed Quantum Computing"

We would like to thank the editor and all three reviewers for their efforts and valuable comments. We have made improvements to the manuscript following comments from reviewers. We highlighted changes in the manuscript in **blue** for the reviewers' convenience. Below we respond in detail to each of the comments from Reviewer 1 to Reviewer 3. All reviewers' comments are reproduced in **italic**.

Response to Reviewer 1's Comments:

1. *In the Introduction, the paper does not clearly distinguish between logical qubits and physical data qubits, which makes the subsequent modeling of qubit allocation and communication ambiguous. Authors are encouraged to provide a section before the system model to cover some important points in quantum basics.*

Response: Thanks for the comment.

[Section 7.B2][1st paragraph] GRDY and EPP are not consider here, since both do not work for Werner systems. For example, both cannot purify a simple 2-hop path with link fidelity at 0.95 to reach an E2E fidelity threshold $F^* = 0.95$, due to their purification order.

2. *The paper introduces CNOT and RCNOT operations but does not provide concrete circuit-level illustrations or explanations of how these operations are implemented. Authors should include what quantum subcircuits are involved and under what conditions multiple operations can be combined or must be treated independently.*

Response: Thanks for the comment.

[Section 7.B2][1st paragraph] GRDY and EPP are not consider here, since both do not work for Werner systems. For example, both cannot purify a simple 2-hop path with link fidelity at 0.95 to reach an E2E fidelity threshold $F^* = 0.95$, due to their purification order.

3. *It is unclear why certain communication operations can be aggregated into a single RCNOT demand while others cannot. In my understanding, entanglements are a must for communication, no matter when using CNOT and RCNOT, and the authors are encouraged to explain the assumptions under which such combinations are valid, and please tell the difference between entanglements generated by (R)CNOT and by swapping.*

Response: Thanks for the comment.

[Section 7.B2][1st paragraph] GRDY and EPP are not consider here, since both do not work for Werner systems. For example, both cannot purify a simple 2-hop path with link fidelity at 0.95 to reach an E2E fidelity threshold $F^* = 0.95$, due to their purification order.

4. *The paper states that the joint qubit allocation and topology design problem is NP-hard, but no proof sketch or reference is provided to justify this claim, which weakens the theoretical foundation of the formulation. Please provide more details about this.*

Response: Thanks for the comment.

[Section 7.B2][1st paragraph] GRDY and EPP are not consider here, since both do not work for Werner systems. For example, both cannot purify a simple 2-hop path with link fidelity at 0.95 to reach an E2E fidelity threshold $F^* = 0.95$, due to their purification order.

5. In Figures 2 and 3, there is an apparent inconsistency, where the qubit interaction graph in Figure 2 indicates interactions between q_4 , q_5 , and q_1 , while Figure 3 later suggests that direct communication between the corresponding QPUs is not possible, and this discrepancy is not explained. Please fix this point.

Response: Thanks for the comment.

[Section 7.B2][1st paragraph] GRDY and EPP are not consider here, since both do not work for Werner systems. For example, both cannot purify a simple 2-hop path with link fidelity at 0.95 to reach an E2E fidelity threshold $F^* = 0.95$, due to their purification order.

6. In Section III-B, the statement “maintaining quantum links in QN is costly, thus we prefer a sparse network topology while maintaining connectivity” is presented as a conclusion without prior explanation or justification, and the logical connection between link cost and sparsity constraints is not formally developed.

Response: Thanks for the comment.

[Section 7.B2][1st paragraph] GRDY and EPP are not consider here, since both do not work for Werner systems. For example, both cannot purify a simple 2-hop path with link fidelity at 0.95 to reach an E2E fidelity threshold $F^* = 0.95$, due to their purification order.

7. The subsequent claim that the optimization therefore includes a constraint on the number of edges appears abrupt, as the paper does not explain why limiting the number of edges is the appropriate abstraction for modeling the link.

Response: Thanks for the comment.

[Section 7.B2][1st paragraph] GRDY and EPP are not consider here, since both do not work for Werner systems. For example, both cannot purify a simple 2-hop path with link fidelity at 0.95 to reach an E2E fidelity threshold $F^* = 0.95$, due to their purification order.

8. The paper discusses a lot of important challenges that need to be solved, such as cost, scheduling, fidelity, and so on. While some of the points are not actually addressed or formulated in the problem statement, the authors are encouraged to review the paper to avoid confusion.

Response: Thanks for the comment.

[Section 7.B2][1st paragraph] GRDY and EPP are not consider here, since both do not work for Werner systems. For example, both cannot purify a simple 2-hop path with link fidelity at 0.95 to reach an E2E fidelity threshold $F^* = 0.95$, due to their purification order.

Response to Reviewer 2’s Comments:

1. The paper addresses my previous questions and proposes a new section on handling large-scale circuits, which is important and valuable. The evaluation part is also almost entirely rewritten: test circuits are replaced with real-world quantum algorithm circuits. Simulated annealing is also added as a new baseline, and QPU configurations have been much more diverse.

Response: Thanks for the comment. We do appreciate your previous questions and suggestions, and happy to see you are satisfied with our extension.

Response to Reviewer 3’s Comments:

1. Edge contraction in the QIG to fit the resource constraints in QPUs could be an interesting new method to partition logical qubits but this should be compared with existing graph partitioning methods to prove its validity.

Response: Thanks for the comment.

[Section 3.B1][1st paragraph] Note that although the process of generating one single pair of entangled photons is probabilistic, our method operates in a time-slotted manner, where the number of successful entanglements

can accumulate at each link, and we use its expectation to represent the available resource on each link. This assumption has been adopted in [10]-[12], [18].

2. *When replacing the CNOT and inter-QPU communications with remote CNOT, explaining why remote CNOT would be more beneficial than using gate teleportation could strengthen the argument of using it. Using hypergraph partitioning and taking advantage of reusing established connections could be considered here. (Also to strengthen the proposed optimization, some other comparisons with existing works that minimize remote operations could be considered.)*

Response: Thanks for the comment. [Section 2.A][related work] [21] has studied stochastic behavior of entanglement swappings and proposed a swapping scheduling method for path-level latency minimization.

[Section 8][future work] (iii) Extending existing results to more accurate system models, such as the stochastic behavior of operations (entanglement generation, swapping [21], and purification) and noisy memory, would also be of interest but cast new technical challenges.

3. *While the qubit interactions and QPU resources are considered, there is the question of how local QPU topologies are constructed and the connectivity of physical qubits. Would the connection and positioning of communication qubits with computation qubits affect remote CNOT operations? If not, this could be mentioned as a reason why this paper will not focus on the local topology of QPUs and instead only focus on the qubit interactions, QPU resources and connections.*

Response: Thanks for the comment. The new results are shown in Fig. 12.

[Section 8][future work] (iii) Extending existing results to more accurate system models, such as the stochastic behavior of operations (entanglement generation, swapping [21], and purification) and noisy memory, would also be of interest but cast new technical challenges.

4. *The swapping success probability should be clarified in the evaluation section and its effects could be noted if significant enough between the different methods. (Since failing the operation would mean an additional retry, this could increase the cost for the remote operation.)*

Response: Thanks for the comment.

5. *In the evaluation section, there should be a clarification of the units for efficiency (objective and time) to make it easier for interpretation.*

Response: Thanks for the comment.

We would like to take this opportunity to express our appreciation for your expertise and invaluable assistance in reviewing our draft. We hope that the revised draft provides the reader with a better presentation of our work. Should you have any questions, please do not hesitate to contact us.

Respectfully yours,

Jiyao, Lei, Yuanxiong, Zhu and Yu